CHETHAN KUMAR C

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PROFESSIONAL SUMMARY

Embedded Firmware Engineer with 2+ years of experience developing and optimizing real-time embedded software across ARM Cortex-based microcontrollers and wireless SoCs (Nordic, Telink, HopeRF). Skilled in C/C++, FreeRTOS, and IoT firmware development, with hands-on experience implementing Device Firmware Update (DFU) mechanisms over UART and OTA, and interfacing sensors via SPI and I2C. Experienced collaborating across firmware, software, and hardware engineering teams to deliver scalable IoT solutions.

PROFESSIONAL EXPERIENCE

Embedded Firmware Engineer | Chargtron IoT Private Limited | Bengaluru, India
July 2024 – July 2026

- Developed and optimized embedded C/C++ firmware for Nordic, Telink, and HopeRF wireless SoCs, delivering reliable functionality across multiple IoT hardware platforms.
- Implemented and validated Device Firmware Update (DFU) mechanisms over UART and OTA, ensuring secure and reliable in-field firmware upgrades.
- Collaborated cross-functionally with software, firmware, and hardware engineering teams to design and deliver complete embedded IoT solutions.
- Built and debugged firmware for ARM Cortex-based microcontrollers using Keil IDE and Eclipse IDE, supporting development from design through testing.
- Interfaced sensors using SPI and I2C bus protocols, integrating sensor data into IoT product firmware.
- Debugged hardware and firmware issues using JTAG, oscilloscope, and logic analyzer, accelerating root-cause analysis during board bring-up and integration testing.
- Tested and validated wireless communication features, including Bluetooth and RF connectivity, across IoT hardware platforms.
- Applied working knowledge of wireless connectivity SoCs (Telink, Nordic, HopeRF) to support firmware architecture and design decisions.

TECHNICAL SKILLS

- **Languages:** C, C++ and Embedded C.
- **Protocols:** UART, I2C, SPI, CAN
- **RTOS:** FreeRTOS
- **Microcontrollers:** ARM Cortex, STM32
- **Network Protocols:** TCP/IP
- **Linux Internals:** Process Management, IPC Mechanisms, File Management, Threads and Memory
- **Tools:** Keil, Eclipse, Segger Embedded Studio, JTAG, Oscilloscope, Logic Analyzer
- **Peripheral Driver Development:** ADC, UART

PROFESSIONAL PROJECTS

Read Analytics Frequency Transmitters (RAFT)

- Designed and developed a wireless feedback system for virtual/classroom education, independently building both transmitter (RAFT) and receiver hardware and firmware using HopeRF microcontrollers.
- Implemented embedded C firmware using 2.4 GHz RF communication with acknowledgment (ACK)-based data transfer, ensuring reliable delivery between transmitter and receiver.
- Designed a 10-button matrix keypad interface on the transmitter to capture and encode student responses.

- Implemented Device Firmware Update (DFU) over OTA for both transmitter and receiver units, enabling in-field firmware upgrades.
- Built a C++ command-line application in Visual Studio to decode, verify, and validate received response data.
- Tested and validated system performance, achieving reliable communication up to 50 feet and support for 50 concurrent responses.

Clicker Screenless Watch

- Developed and tested a screenless wearable health-monitoring device built on a HopeRF microcontroller, featuring 2.4 GHz RF and BLE connectivity for pairing with Android mobile devices.
- Interfaced the BMA400 sensor via SPI to data for step counting, tap detection, and fall detection.
- Implemented a panic-press feature to trigger alerts via BLE to the connected Android app.
- Designed tap detection and fall detection logic using accelerometer data for gesture-based, screenless user interaction.
- Designed LED-based feedback to communicate device status in the absence of a display.
- Validated step-count accuracy against Google Fit in real-time testing, achieving up to 96% accuracy; estimated battery life of 6+ months under daily usage.

EDUCATION

B.Tech in Electronics and Communication Engineering (ECE)
Hindustan Institute of Technology and Science (Hindustan University) | 2019 – 2023 | Aggregate: 83%

Higher Secondary (Class XII)
Sri Vijay Vidyalaya Matriculation Higher Secondary School (State Board) | 2017 – 2019 | Aggregate: 67%

Secondary School Leaving Certificate (SSLC)
Ineca Matriculation School (State Board) | 2016 – 2017 | Aggregate: 90%

PROFESSIONAL TRAINING

Embedded Systems Training, Vector India Pvt. Ltd., Bengaluru | July 2023 – March 2024

PERSONAL DETAILS

- **Date of Birth:** 28/11/2001
- **Gender:** Male
- **Languages:** English, Telugu, Tamil, Kannada
- **Address:** 4/67, Sathanur (Village), Krishnagiri (District), Tamil Nadu – 635114

DECLARATION

I hereby declare that all the above information is true to the best of my knowledge and belief.

Place: Hosur

CHETHAN KUMAR